

Jiantao Liu

☎ +41 79 529 7047 | @ jianliu@ethz.ch | 🔗 LinkedIn | 🌐 Website | 📍 Zürich, Switzerland

EDUCATION

ETH Zurich

Ph.D. in Electrical Engineering and Information Technology;

Research focus: Computer Architecture, Electronic Design Automation, Efficient Computing System

Advisor: Prof. Dr. Lana Josipović

Zurich, Switzerland

Oct 2023 – Present

University of California San Diego

Ph.D. in Computer Engineering; GPA: 4.0/4.0

San Diego, U.S.A.

Sep 2022 – Jun 2023

ETH Zurich

M.Sc. in Electrical Engineering and Information Technology; GPA: 5.25/6.00

Zurich, Switzerland

Sep 2020 – Sep 2022

Huazhong University of Science and Technology

B.Eng. in Electrical and Electronics Engineering; GPA: 3.92/4.00

Wuhan, China

Sep 2016 – Jun 2020

WORK EXPERIENCE

Huawei Hong Kong Research Center

Research Intern, CPU Architecture Team

Hong Kong, China

April 2026 – Oct 2026

- Conduct system performance engineering for next-generation CPU architectures, with a focus on computational-kernel analysis and architectural evaluation.
- Evaluate the impact of new Arm ISA extensions and Scalable Matrix Extension (SME) on computational kernels, and analyze agentic AI workloads to identify performance opportunities for future CPUs.

PUBLICATIONS

1. **Jiantao Liu**, Carmine Rizzi, and Lana Josipović. *Load-Store Queue Sizing for Efficient Dataflow Circuits*. In Proceedings of the IEEE International Conference on Field-Programmable Technology (FPT), Hong Kong, China, Dec. 2022.
2. Matteo Perotti, Matheus A. Cavalcante, Alessandro Ottaviano, **Jiantao Liu**, and Luca Benini. *Yun: An Open-Source, 64-Bit RISC-V-Based Vector Processor With Multi-Precision Integer and Floating-Point Support in 65-nm CMOS*. *IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II)*, 2023.
3. **Jiantao Liu**, Maksymilian Graczyk, Andrea Guerrieri, and Lana Josipović. *Fast Switching Activity Estimation for HLS-Produced Dataflow Circuits*. In Proceedings of the IEEE International Conference on Field-Programmable Logic and Applications (FPL), Torino, Italy, Sep. 2024.
4. **Jiantao Liu**, Minxuan Zhou, Yue Pan, Chien-Yi Yang, Lana Josipović, and Tajana Rosing. *OptiPIM: Optimizing Processing In-Memory Acceleration Using Integer Linear Programming*. In Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA), Tokyo, Japan, Jun. 2025. **Distinguished Artifact Award**.
5. **Jiantao Liu**, Carmine Rizzi, Pravriti Jaipuriyar, Ulrike Schorr, Sascha Richter, Ankush Sood, and Lana Josipović. *GLOW: Glitch Optimization with Retiming for Low Power*. In Proceedings of the International Workshop on Logic and Synthesis (IWLS), Hong Kong, China, Jun. 2026. **Best Student Paper Award**.
6. Chien-Yi Yang*, **Jiantao Liu***, Yue Pan, Tajana Rosing, Minxuan Zhou, and Lana Josipović. *PNMAX: Mapping-Architecture Co-Exploration Framework for Processing Near Memory*. In Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture (MICRO), Athens, Greece, Oct. 2026. *To appear*.
7. **Jiantao Liu**, Carmine Rizzi, Pravriti Jaipuriyar, Ulrike Schorr, Sascha Richter, Ankush Sood, and Lana Josipović. *GLOW: Glitch Optimization with Retiming for Low Power*. In Proceedings of the IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, USA, Nov. 2026. *To appear*.

*Equal contribution (co-first authorship).

TAPEOUT & OPEN-SOURCE PROJECTS

- **Yun, Vector Processor**, TSMC 65 nm (tape-out 2021). First silicon implementation of Ara, the RISC-V Vector extension engine for the CVA6 (Ariane) core. Responsible for the backend implementation process (SP&R flow and DFT).
- Developer of **Dynatonic**, an open-source HLS compiler based on MLIR that produces dynamically scheduled, dataflow circuits out of C/C++ code.

TEACHING EXPERIENCE & PROFESSIONAL SERVICE

- Teaching Assistant · *Synthesis of Digital Circuits* (Spring 2024), ETH Zürich.
- Journal Reviewer: IEEE TCAD, IEEE TPDS, ACM TODAES.

SKILLS

Programming & HDLs: C/C++, Python, OCaml, SQL, P4, Scala, System Verilog, Chisel, HTML, LabView, PyTorch, TensorFlow

Tools: LLVM, MLIR, Synopsys Design Compiler, Fusion Compiler, Vivado, Vitis, Cadence Genus, Innovus, Joules, Calibre, OpenRoad, MySQL, MongoDB, Docker, OpenCV, Cerebras SDK